



*Warmth for Every Journey*

# NOVARA MATERIALS

## Direct-Contact Battery Heating Technology

*A Technical White Paper for Automotive Tier 1 & OEM Partners*

June 2026 | NovaraMaterials.com

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### Executive Summary

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Battery electric vehicles (BEVs) face a fundamental thermal challenge that coolant-based heating architectures were never designed to solve efficiently. When ambient temperatures drop below 0°C, conventional high-voltage coolant heater (HVCH) systems introduce compounding energy losses through thermal mass, pumping overhead, and multi-stage heat transfer — resulting in +30% range loss, charging times that triple, and accelerated battery aging that can shorten pack life by four to eight years.

Novara Materials has developed a dual use product that is a dielectric barrier and a direct-contact thin film heating technology based on proprietary carbon nanotube (CNT) coatings that eliminates these inefficiencies at the source. By applying resistive heating directly to battery cells, cooling ribbons, or cold plates — rather than circulating a heated fluid — Novara's system delivers heat where it is needed with 56% less energy, 81% faster warm-up, and 62% faster charging versus the coolant-based benchmark.

This white paper presents the technical basis for Novara's performance claims, the commercial opportunity for Tier 1 suppliers and OEMs, and the specific application path for integrating Novara's film as a dual-use dielectric barrier and heater in existing battery thermal management systems (BTMS).

## 1. The Cold-Weather BEV Problem

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### 1.1 Heat Is No Longer Free

Internal combustion engine vehicles generate abundant waste heat as a byproduct of combustion — cabin and battery heating in a conventional vehicle is essentially free. BEVs produce no such waste heat. Every joule used to warm the battery or cabin must be drawn from the battery pack, directly reducing available range. In cold climates this creates a compounding penalty: the battery needs more heat to function well, yet heating it costs the very energy it is meant to store.

### 1.2 Cold-Weather Performance Degradation

Peer-reviewed research and independent real-world testing quantify the magnitude of the problem:

- 31% range loss at -7°C observed across multiple production BEV platforms including Hyundai Ioniq 6, Kia EV6, Tesla Model 3, and Nissan Leaf (AAA / Recurrent, 2023–2024).
- 3–5x longer charging times at -20°C; constant-current phase delivers only 14% of normal charging capacity at -30°C.

- 40% capacity loss in fewer than 100 cycles at 0°C with 2C charging — effectively destroying a battery pack in less than one winter season of aggressive use (You, Dai & Li, SAE 2019-01-1204).
- Internal resistance increases 4.5x at -25°C versus the 25°C baseline, reducing both peak power delivery and charging acceptance.

### 1.3 Safety: The Lithium Plating Mechanism

Cold-weather charging does not merely degrade performance — it creates genuine safety hazards. When lithium-ion cells are charged below 0°C, lithium ions fail to intercalate cleanly into the graphite anode and instead deposit as metallic dendrites on the anode surface. Post-mortem analysis of cold-cycled cells reveals a concrete-like  $\text{Li}_2\text{CO}_3$  layer covering the anode and visible metallic lithium plating (Spettmann et al., SAE 2024-01-2063). These dendrites:

- Permanently reduce capacity as lithium becomes electrically isolated from the circuit.
- Create internal short-circuit pathways capable of piercing the separator.
- Dramatically increase thermal runaway risk — aged pouch cells subjected to mechanical intrusion reached 120°C after internal short circuit, only 30°C below typical runaway initiation temperature.

### 1.4 Warranty Exposure

Cold-weather cycling is the dominant cause of premature battery aging across NMC chemistries — the most common chemistry in current BEV production. Without effective thermal management, batteries in cold-climate markets may require replacement 4–5 years earlier than the nominal 8–10 year lifecycle. At \$8,000–\$15,000 per replacement, a 10% premature failure rate across a 500,000-unit annual production program represents up to \$270 million in warranty exposure over a 10-year period.

## 2. Novara's Direct-Contact Heating Technology

### 2.1 The CNT Thin Film Difference

Novara's heaters are manufactured by applying a proprietary carbon nanotube conductive coating to polymer film or non-woven substrate via a roll-to-roll process. The resulting film is 0.28 mm thick, weighs 0.037 g/cm<sup>2</sup>, and functions as a resistance heater with uniform heat distribution across its entire surface area — unlike wire-based elements that create concentrated hot zones at each wire pass.

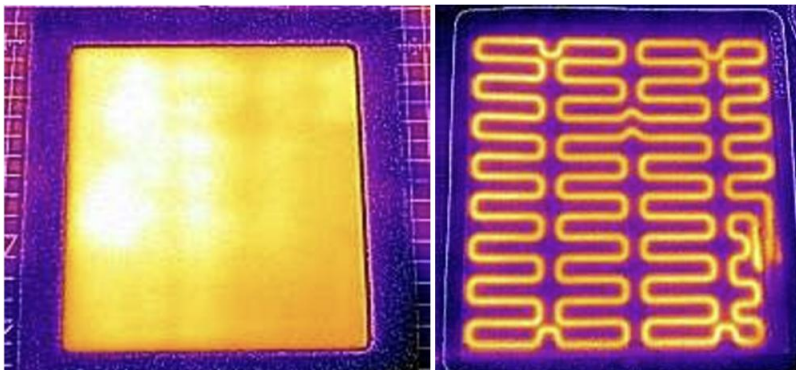


Figure 1: Novara's CNT heater vs. Wire Heater

The critical distinction from coolant-based systems is the elimination of all intermediate heat transfer steps. A conventional HVCH must heat coolant fluid, pump that fluid through a loop, transfer heat through a cold plate interface, and then conduct through the battery casing before reaching cells. Each step introduces thermal resistance, pumping losses, and time delay. Novara's film is placed in direct contact with the cell surface, cooling ribbon, or cold plate baseplate — transferring heat in one step.

2.2 Measured Performance vs. Coolant Baseline

| Key Metric                               | Coolant Baseline | Novara Direct-Contact | Improvement |
|------------------------------------------|------------------|-----------------------|-------------|
| 10-80% SOC Charging Time (-20°C to 20°C) | 74 min           | 28 min                | -62%        |
| Warm-Up Time (-20°C to 20°C)             | 58 min           | 11 min                | -81%        |
| Energy Required for Heating              | 3.8 kWh          | 1.7 kWh               | -56%        |
| Winter EV Range                          | 312 km           | 388 km                | +24%        |

Table 1: Direct-contact film heating versus coolant-based baseline (-20°C to 20°C test conditions)

The warm-up test result of 513 seconds (8.5 minutes) from -20°C to 20°C compared to a benchmark of approximately 4,000 seconds represents a 7.8x speed improvement — achieved while consuming 56% less energy. This efficiency gain is attributable directly to the elimination of thermal mass losses and pumping overhead inherent in coolant loop architectures.

2.3 Manufacturing Scalability

Novara operates a 50,000 ft² cleanroom manufacturing facility in Ohio, USA, equipped with roll-to-roll coating and converting lines capable of producing 72" wide film at 600 ft²/minute. This throughput enables high-volume automotive supply without the tooling complexity or capital intensity associated with injection-molded or stamped thermal components. The roll-to-roll format also enables tight dimensional control and in-line quality inspection across the full film width.

### 3. System Integration: Dual-Use Dielectric & Heater

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#### 3.1 Replacing the Dielectric Coating

The most compelling near-term integration path for Tier 1 battery system suppliers is the replacement of existing dielectric coatings on cooling ribbons, cold plates, and cell interfaces. These coatings currently serve a single purpose — electrical isolation — and represent a cost center with no performance upside. Novara's film replaces this coating while adding resistive heating capability, effectively converting a passive insulator into an active thermal management component at minimal incremental cost and no additional packaging volume.

#### 3.2 HVCH Elimination & System-Level Benefits

Beyond the ribbon-level performance improvement, integrating Novara's film into the BTMS architecture enables elimination of the high-voltage coolant heater (HVCH) — a significant system-level simplification. Current HVCH implementations require a dedicated high-voltage heating element, coolant circuit, manifolds, pump, and associated plumbing and connectors. Removing this subsystem delivers:

- Packaging volume recovery in a space-constrained battery pack envelope.
- Reduction in component count, connector interfaces, and associated failure modes.
- Weight savings from elimination of coolant heater, pumps, manifolds, and associated coolant mass.
- Simplified thermal management controller architecture with direct resistive heating control.

#### 3.3 Application Matrix

Novara's thin film is compatible with three primary integration configurations in current BEV battery pack architectures:

- **Slithering ribbon format for cylindrical module assemblies.**
- **Baseplate cold plates for pouch or prismatic module assemblies.**
- **Straight ribbon format for prismatic module assemblies.**



## 4. Commercial Value for OEM and Tier 1 Partners

### 4.1 Customer Value Created

A value-based analysis quantifies the economic benefit Novara's technology delivers over a 10-year vehicle ownership period:

| Value Component               | Amount           | Explanation                                                |
|-------------------------------|------------------|------------------------------------------------------------|
| Battery replacement avoidance | ~\$10,000        | Prevents cold-weather lithium plating & premature aging    |
| Driver time savings           | ~\$7,500         | 47 min saved per charge × \$20/hr × 50 charges/yr × 10 yrs |
| Winter range extension value  | ~\$3,000         | 76 km additional range (~5-10% of vehicle purchase price)  |
| Enhanced resale value         | ~\$2,000         | Cold-weather capability premium in used EV market          |
| <b>Total Customer Value</b>   | <b>~\$22,500</b> | <b>Over 10-year vehicle ownership</b>                      |
|                               |                  |                                                            |

Table 2: Customer lifetime value analysis — Novara direct-contact battery heating system

### 4.2 Consumer Demand Validation

A 107-respondent consumer survey conducted in February 2026 validated market demand for cold-weather BEV performance improvements:

- 70.76% rated cold-weather performance as 'Critical' or 'Very Important' in EV purchase decisions.
- 54.21% expressed willingness to pay \$251–\$2,000+ for a technology delivering the described benefits.
- 67.29% would prefer or strongly prefer brands offering this capability over those that do not.
- 73.83% requested more information about battery longevity benefits — indicating a significant education-driven upsell opportunity.

### 4.3 OEM Partner Financial Impact

For a Tier 1 supplier or OEM with 500,000 annual BEV units, the financial opportunity is substantial:

- \$50M–\$195M annual incremental revenue at 25%–60% attachment rates
- \$216M in warranty cost avoidance over 10 years by reducing cold-climate premature battery failures from an estimated 10% to the normal ~2% rate.
- 5–10% market share shift potential in cold-climate states where 67% of consumers prefer brands with this capability.

## 5. Technical Specifications

| Parameter                 | Specification                                      |
|---------------------------|----------------------------------------------------|
| Thickness                 | 0.28 mm (target: 0.20 mm for ribbon apps)          |
| Weight                    | 0.037 g/cm <sup>2</sup>                            |
| Thermal Conductivity      | Customizable                                       |
| Dielectric Strength       | Customizable                                       |
| Voltage Range             | Customizable - 6 VDC – 480 VAC                     |
| Max Operating Temperature | Customizable                                       |
| Power Density             | Customizable                                       |
| Manufacturing Format      | Roll-to-roll, 72" wide at 600 ft <sup>2</sup> /min |
| Substrate Compatibility   | Film or non-woven; adhesive-backed available       |

Table 3: Novara thin film heater technical specifications — current and development targets



Figure 2: Roll to Roll Format

## 6. Development Roadmap & Next Steps

Novara is actively engaged with Tier 1 BTMS suppliers in thermal and mechanical qualification testing. The current development priorities for ribbon and cold plate applications are:

- Thermal conductivity enhancement through filler loading and substrate optimization — targeting completion Q3 2026.
- Thickness reduction to 200–220 µm while maintaining dielectric and conductivity requirements.
- Glycol compatibility validation per automotive environmental test standards (alternating temperature, condensation/humidity, pulse pressure).

- Module-level cold chamber testing in collaboration with Tier 1 partner — supply of uncoated ribbons and cells for integrated validation.
- AEC-Q200 and IATF 16949 certification program initiation.

Novara welcomes engagement from Tier 1 suppliers and OEM thermal and battery engineering teams for joint development, sample evaluation, and pilot program structuring. A partnership pathway is available for early-mover partners seeking to establish brand leadership in cold-weather EV performance.

## References

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